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Lab Work 09: Observer Pattern

In a futuristic city filled with neon lights, there is a central computer system called the **`Bounty Dispatcher`**. This system acts as the boss that controls all the high-stakes missions in the city. The Dispatcher is the central hub that knows exactly when a new criminal has been spotted or a new job is available.

Out in the city, there are many different **`Bounty Hunters`** waiting for work. Some of these hunters are high-tech hackers called **`Netrunners`** and others are skilled warriors called **`Street Samurais`**. These hunters do not want to sit around and check a website every five minutes to see if there is a new job. Instead, they connect their neural links to the **`Bounty Dispatcher`** so they can receive instant alerts.

As soon as a new bounty is entered into the system, the Dispatcher automatically sends a signal to every single hunter on its list. The **`Netrunners`** might see the data and prepare their hacking tools, while the **`Street Samurais`** might see the same data and grab their swords. Each hunter reacts differently to the news, but the Dispatcher sends the exact same message to everyone at the same time.

Create a central **`Bounty Dispatcher`** hub and connect to that hub with two **`Bounty Hunters`**.(One Samurai and one Netrunner)

Sometimes a hunter might get injured or decide that the job is too dangerous. When this happens, they can choose to disconnect their neural link from the network. The Dispatcher then removes them from the secret list immediately.

To simulate that situation, assume that your **`Netrunner`** got injured and will no longer be able to take any jobs.

At some points in a day **`Bounty Dispatcher`** hub updates itself to receive information from the satellite but for only that process, **`Bounty Hunters`** don't need to know the information hub gets from the satellite.